

MANGALORE, Australia

WMO: 948740

Lat: 36.8886S

Lon: 145.1858E

Elev: 464

StdP: 14.45

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.6	33.6	28.1	22.4	57.4	30.4	24.9	50.9	22.7	53.3	20.7	52.2	2.7	210	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.3	97.1	65.5	92.8	65.3	89.1	64.4	71.5	82.4	69.1	82.6	67.2	82.0	12.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	108.4	74.4	65.5	95.6	72.2	62.7	86.8	71.0	35.6	81.7	33.5	82.4	31.9	81.9	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.4	20.2	18.3	DB	27.9	104.4	1.6	3.6	26.8	107.0	25.8	109.2	25.0	111.2	23.8	113.9
			WB	27.9	78.1	1.7	6.5	26.7	82.8	25.7	86.6	24.7	90.3	23.5	95.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	71.9	71.1	66.2	58.9	52.4	47.9	46.6	47.8	51.5	56.8	63.2	67.3
	DBStd	10.57	7.19	6.54	6.42	5.52	4.58	3.96	3.61	3.95	4.80	6.00	6.88	7.09
	HDD50	366	0	0	0	3	26	87	116	91	35	7	1	0
	HDD65	3137	17	18	61	194	390	514	571	533	405	265	114	55
	CDD50	3431	680	590	504	270	101	22	10	23	80	219	396	536
	CDD65	729	232	188	99	12	0	0	0	0	1	11	60	126
	CDH74	9229	2914	2139	1095	142	1	0	0	0	11	256	903	1768
	CDH80	4171	1507	994	409	19	0	0	0	0	1	75	353	813
Wind	WSAvg	9.0	11.2	10.4	9.5	8.0	7.4	7.3	7.6	7.9	8.4	9.3	10.2	10.7
Precipitation	PrecAvg	22.6	1.5	1.3	1.2	1.6	2.3	2.3	2.4	2.6	2.3	2.0	1.8	1.6
	PrecMax	38.3	4.6	8.5	4.3	6.5	5.3	6.7	4.7	5.9	7.0	6.9	5.9	4.8
	PrecMin	10.2	0.0	0.0	0.1	0.2	0.2	0.5	0.3	0.4	0.5	0.1	0.0	0.0
	PrecStd	5.6	1.1	1.7	0.9	1.2	1.4	1.3	1.3	1.3	1.5	1.4	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.0	100.8	95.4	82.5	71.1	62.9	60.0	65.5	74.7	87.7	94.2	100.5
		MCWB	67.3	68.3	65.1	60.6	56.4	53.0	52.1	55.8	57.8	61.6	63.1	65.3
	2%	DB	98.1	94.4	88.9	77.9	66.8	59.3	57.0	61.3	69.3	81.0	88.7	93.8
		MCWB	66.2	66.5	64.0	60.2	56.0	52.7	50.8	52.9	57.3	59.0	62.3	64.2
	5%	DB	93.0	90.2	84.5	74.5	63.7	57.3	55.2	58.6	65.6	75.5	83.7	88.8
		MCWB	65.8	65.8	63.4	58.9	54.9	51.6	49.8	51.2	55.6	58.1	61.5	63.0
	10%	DB	88.4	86.3	80.2	70.9	61.1	55.4	53.6	56.4	62.4	70.7	78.8	84.0
		MCWB	64.8	64.7	62.0	57.8	54.0	50.5	48.9	49.9	53.9	56.6	60.5	62.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.7	76.5	71.6	66.5	60.8	56.6	54.8	56.9	62.2	65.4	69.8	74.1
		MCDB	82.6	82.6	76.7	74.7	64.5	58.7	56.9	63.1	67.4	78.2	82.4	79.9
	2%	WB	71.2	71.1	68.0	62.8	58.2	54.2	52.3	54.1	59.0	62.1	66.3	68.1
		MCDB	84.8	82.1	80.3	70.5	62.9	57.4	55.2	59.7	65.6	72.7	79.3	81.2
	5%	WB	68.6	68.8	65.7	60.8	56.4	52.6	50.8	52.2	56.6	60.0	64.1	65.8
		MCDB	85.0	82.4	78.3	69.9	61.6	56.1	53.9	57.5	64.2	71.2	77.0	80.7
	10%	WB	66.6	66.8	63.8	58.9	54.9	51.2	49.5	50.5	54.4	57.8	62.1	63.8
		MCDB	84.3	81.5	76.4	69.4	60.3	54.7	52.6	55.5	61.2	69.3	74.5	80.2
Mean Daily Temperature Range	5% DB	MDBR	26.3	24.9	23.0	21.3	17.4	15.0	14.3	16.8	20.1	23.3	24.3	26.1
		MCDBR	34.7	32.1	30.1	27.2	20.7	17.3	16.3	20.3	25.6	32.9	33.3	34.7
		MCWBR	10.3	9.8	10.2	12.3	11.5	11.0	10.8	12.5	14.8	14.5	11.4	11.6
	5% WB	MCDBR	27.4	25.0	23.7	22.2	16.9	13.8	13.7	18.3	22.1	26.2	25.1	26.9
		MCWBR	11.3	11.3	11.1	12.2	10.4	9.9	9.8	12.1	14.0	14.1	11.6	12.4
Clear-Sky Solar Irradiance	taub	0.344	0.344	0.321	0.314	0.301	0.293	0.291	0.301	0.326	0.329	0.342	0.335	
	taud	2.494	2.503	2.564	2.552	2.567	2.581	2.585	2.541	2.460	2.476	2.474	2.511	
	Ebn at Noon	314	306	302	285	270	264	272	285	295	308	312	318	
	Edh at Noon	36	34	30	26	23	21	22	26	32	35	36	36	
All-Sky Solar Radiation	RadAvg	2388	2101	1704	1202	806	616	672	955	1357	1808	2120	2364	
	RadStd	134	113	84	63	53	66	39	73	112	149	115	117	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)		+0.92	N/A	N/A	+1.73	N/A	N/A	-32	-232	+333	+119

Nomenclature: See separate page